

# OceanCyber POS

## Development Guide

[Complete multi-tenant SaaS specification](#)

Self-register · BYOK payments · Offline & manual MoMo

Subscription billing · Onboarding wizard · API & schema

Ghana market · MoMo · Multi-branch · Separate product domain

# Table of contents

1. Product vision & positioning
2. Scope boundary
- A. Complete SaaS model (overview)
- B. Self-service registration flow
- C. Onboarding wizard (5 steps)
- D. Merchant payment configuration (BYOK)
- E. Offline payment modes
- F. SaaS subscription billing (OceanCyber!' merchant)
- G. Database — SaaS auth & payment config
- H. REST API — registration & onboarding
- I. REST API — payment settings
- J. Payment UI — POS checkout screen
- K. Security — tenant payment secrets
- L. Marketing site integration (SaaS)
- M. SaaS sprint backlog additions
- N. Revised sales paths (dual channel)
- O. Open decisions — updated (v4.0)
- 3.1 Marketing spec — checkout
- 3.2 Marketing spec — offline sync
- 3.3 Marketing spec — inventory & branches
- 3.4 Marketing spec — restaurant & onboarding
4. Industry vertical presets
5. Pricing tiers & feature gates
6. Recommended tech stack
7. Repository layout (proposed)
8. System architecture
9. Database schema — core tenancy
9. Database schema — catalog & inventory
9. Database schema — sales & payments
9. Database schema — restaurant & sync
10. REST API — auth & tenancy
10. REST API — catalog & sync
10. REST API — sales & payments
10. REST API — inventory, reports, admin
11. WebSocket events (live dashboard)
12. MoMo payment state machine
13. Offline sync protocol (detail)
14. UI screens — POS desktop client
14. UI screens — admin portal
15. Security requirements
16. Environment variables (product)
17. Local development setup
18. Deployment — Coolify VPS
19. Phase 0 sprint backlog (weeks 1–6)
20. Phase 1 MVP backlog (weeks 7–16)
21. Phase 2 growth backlog (weeks 17–24)
22. Phase 3 enterprise backlog
23. Testing strategy
24. GRA E-VAT & compliance roadmap
25. Acquisition & onboarding paths
26. Marketing!' engineering alignment
27. Competitive differentiation (Ghana 2026)
28. Open decisions (log before build)
29. Ghana tax calculation (retail defaults)
30. Permission matrix (RBAC)
31. API error codes

- 32. Sample JSON — create sale (online)
- 33. Sample JSON — sync push (offline batch)
- 34. Sample JSON — MoMo initiate & webhook
- 35. Receipt template (80mm thermal)
- 36. SMS receipt template
- 37. Hardware kit specifications
- 38. Supported devices (certify before sale)
- 39. Entity relationships (ERD summary)
- 40. Database indexes (performance)
- 41. Caching & rate limits
- 42. CI/CD pipeline
- 43. Versioning & release policy
- 44. Electron auto-update flow
- 45. Backup & disaster recovery
- 46. Monitoring & alerting
- 47. Performance targets (SLA)
- 48. Catalog CSV import format
- 49. Merchant onboarding checklist
- 50. OceanCyber internal provision checklist
- 51. Marketing site !' SaaS app integration
- 52. Dual billing architecture
- 53. Branch transfer workflow
- 54. Restaurant order lifecycle
- 55. Shift reconciliation procedure
- 56. Void & refund policy matrix
- 57. Fraud & abuse controls
- 58. Support tiers & SLA
- 59. Incident response runbook
- 60. Data retention schedule
- 61. Penetration test scope (pre-enterprise)
- 62. Team roles & ownership
- 63. Budget & timeline summary
- 64. Glossary
- 65. Document revision history
- P. Wireframe — Sign up & sign in
- P. Wireframe — Onboarding wizard (steps 1–3)
- P. Wireframe — Onboarding payments (step 4)
- P. Wireframe — Settings !' Payments (HQ admin)
- P. Wireframe — POS checkout payment drawer
- Q. Prisma schema — tenancy & SaaS billing
- Q. Prisma schema — users & payment providers
- Q. Prisma schema — sales & payments (SaaS fields)
- R. OpenAPI — auth & registration (v1)
- R. OpenAPI — onboarding & billing
- R. OpenAPI — payment settings & checkout
- S. Sample JSON — self-serve registration
- S. Sample JSON — save Paystack BYOK config
- S. Sample JSON — manual MoMo payment
- T. Tenant lifecycle state machine
- U. Email templates (transactional)
- V. 16-week delivery plan (story points)
- W. Environment variables (v5 complete list)
- X. QA checklist — SaaS signup to first sale
- Y. Document revision — v5.0

## 1. Product vision & positioning

OceanCyber POS is a **complete multi-tenant SaaS** for Ghanaian operators: self-service registration, subscription billing, configurable payments (merchant's own Paystack/Hubtel keys or offline/cash), offline-first POS, and multi-branch control.

Market anchor: from GHS 99/month (SellarPro, CliqPOS, Retill, Konect benchmark).

Premium edge: optional assisted onboarding for Enterprise — self-serve is the default path.

Dual acquisition: Start free trial (self-serve) OR Book demo (sales-assisted) from marketing site.

Currency: GHS only in UI and reports; OceanCyber subscription billed separately from in-store customer payments.

Product app: app.pos.oceancyber.net — separate from oceancyber.net marketing site.

## 2. Scope boundary

IN SCOPE (POS SaaS product):

- Self-service signup, email verification, onboarding wizard
- SaaS subscription billing (OceanCyber Paystack)
- Merchant BYOK payment config (Paystack, Hubtel) OR offline/manual MoMo + cash
- Electron/tablet clients, admin portal, inventory, multi-branch, restaurant module

Marketing site (oceancyber.net): product pages linking to app.pos.oceancyber.net/signup — no POS app hosted there.

OUT OF SCOPE on marketing site: POS login UI, tenant admin, payment key storage.

### A. Complete SaaS model (overview)

OceanCyber POS is a **full multi-tenant SaaS product**: merchants **self-register**, complete an onboarding wizard, choose a plan, and configure their own environment — including **their own payment providers** or **offline-only** mode.

Two payment layers (never confuse them):

1. **SaaS subscription** — merchant pays OceanCyber (GHS 99+/mo) via Paystack for using the software.
2. **In-store payments** — merchant's customers pay the merchant via cash, offline MoMo, or merchant-configured Paystack/Hubtel/telco keys.

OceanCyber does NOT take a cut of in-store MoMo transactions unless explicitly offered as a future aggregator tier.

Premium **assisted onboarding** remains available for Enterprise — but is optional, not required to start.

### B. Self-service registration flow

Public URLs (POS product domain, not marketing site):

- app.pos.oceancyber.net/signup — create account
- app.pos.oceancyber.net/signin — returning users
- app.pos.oceancyber.net/onboarding — post-signup wizard (auth required)

Registration fields:

- Full name, work email, phone (233...), password (min 12 chars)
- Business / trading name
- Accept Terms of Service + Privacy Policy + DPA processor clause

Flow:

1. `POST /api/v1/auth/register !'` create user + tenant (
2. Send verification email (magic link or 6-digit code, 15 min TTL)
3. `GET /api/v1/auth/verify-email?token=... !'` activate tenant (status: trialing or ac
4. Redirect to onboarding wizard (5 steps)
5. On wizard complete !' tenant status: active, first branch + admin user ready

Rate limit: 5 registrations/hour per IP; captcha (hCaptcha/Turnstile) on signup.

### C. Onboarding wizard (5 steps)

Step 1 — Business profile: legal name, TIN (optional), industry preset, timezone Africa/Accra.

Step 2 — First branch: name, address, phone, receipt footer text.

Step 3 — Plan selection: Starter / Growth / Multi-branch (14-day trial on Starter optional).

Step 4 — Payment setup (see section D): configure providers OR offline-only.

Step 5 — Invite team + pair first device (QR pairing code, 10 min expiry).

Skip allowed: Steps 4–5 skippable — default to **cash-only offline mode** until merchant configures payments later in Settings.

Progress saved per step; resume on re-login.

Empty catalog: offer sample products import or CSV upload template download.

## D. Merchant payment configuration (BYOK)

Merchants configure **their own** payment credentials in Settings! Payments. OceanCyber stores secrets **encrypted** (AES-256-GCM, tenant-specific data key in KMS/env).

Never return secret keys to client after save — only masked preview (pk\_live\_...xxxx).

Supported provider modules (enable/disable per tenant):

- **Cash** — always available
- **Offline MoMo** — record MoMo payment manually (reference number, no API) — see section E
- **Paystack** — merchant's public + secret key; subaccount optional
- **Hubtel** — client id + secret + merchant account number
- **Direct telco** — Enterprise roadmap (MTN/Telecel API credentials)
- **Card terminal** — phase 2 hardware adapter

Test connection button: POST /api/v1/settings/payments/test — validates keys against provider sandbox/live.

Webhook URLs shown in UI for merchant to paste into Paystack/Hubtel dashboard:

```
https://api.pos.oceancyber.net/api/v1/webhooks/paystack/{tenant_slug}
https://api.pos.oceancyber.net/api/v1/webhooks/hubtel/{tenant_slug}
```

## E. Offline payment modes

Three distinct modes — implement all:

**E1 — Cash only (default after signup skip)**

No external payment API. Cash tender + change calculation. Full offline sync supported.

**E2 — Offline MoMo (manual recording)**

Cashier selects MoMo! customer pays on phone outside system! cashier enters:

- Provider (MTN / Telecel / AT),
- Amount,
- External reference (optional),
- MSISDN last 4 digits

No OTP/API call. Receipt shows 'MoMo — manual confirmation'. Reconciliation report for finance.

Ideal for merchants without Paystack/Hubtel or when network/API down (fallback prompt).

**E3 — Offline-first POS operations**

Device loses internet: all payment modes above queue locally.

Automated MoMo (API) disabled offline with clear UI message; manual MoMo + cash still work.

On reconnect: sync sales; automated MoMo payments cannot be initiated retroactively for queued sales — cashier completes or converts to manual MoMo.

Tenant setting: payments.default\_mode = cash | manual\_momo | automated (requires configured provider).

## F. SaaS subscription billing (OceanCyber! merchant)

Separate from in-store payments. Billed to tenant owner email via **OceanCyber Paystack account**.

Plans map to Paystack Plan codes (OceanCyber-owned, not merchant keys):

- pos\_starter\_monthly — GHS 99
- pos\_growth\_monthly — GHS 199
- pos\_multibranch\_monthly — GHS 299

Trial: 14 days Starter, card/MoMo optional on signup (Paystack subscription with trial\_period\_days).

Without payment method at trial end: tenant! read\_only (view reports, no new sa

API:

POST /api/v1/billing/subscribe — { plan\_code, payment\_method } (Paystack authorization)

GET /api/v1/billing/status — plan, renews\_at, trial\_ends\_at

POST /api/v1/billing/webhooks/paystack — OceanCyber subscription events (separate webhook secret)

Enforce plan limits in middleware: branches, users, devices, feature flags.

## G. Database — SaaS auth & payment config

tenants — add: status (pending|trialing|active|past\_due|suspended|cancelled),  
 onboarding\_step, trial\_ends\_at, plan\_code, paystack\_customer\_id, paystack\_sub\_code

users — add: email\_verified\_at, verification\_token\_hash, password\_reset\_token\_hash

tenant\_payment\_providers — id, tenant\_id, provider (cash|manual\_momo|paystack|hubtel),  
 is\_enabled, config\_encrypted (JSON blob), webhook\_secret\_encrypted,  
 last\_tested\_at, test\_status, created\_by

tenant\_payment\_settings — tenant\_id, default\_mode, allow\_manual\_momo\_fallback,  
 require\_supervisor\_for\_manual\_momo\_above\_ghs (default 500)

payments — add column: capture\_type (automated|manual|cash), external\_ref\_manual

audit\_payment\_config\_changes — tenant\_id, user\_id, provider, action, ip, at

## H. REST API — registration & onboarding

POST /api/v1/auth/register  
 { name, email, phone, password, business\_name, accept\_terms: true }  
 !' 201 { user\_id, tenant\_id, verification\_sent: true }

POST /api/v1/auth/verify-email { token }

POST /api/v1/auth/resend-verification

GET /api/v1/onboarding — current step + saved data

PATCH /api/v1/onboarding/step/{1-5} — partial save

POST /api/v1/onboarding/complete — validate all required fields

POST /api/v1/billing/start-trial — optional at step 3

POST /api/v1/billing/subscribe — Paystack checkout or inline authorization

## I. REST API — payment settings

GET /api/v1/settings/payments — enabled providers (secrets masked)

PUT /api/v1/settings/payments/{provider} — save encrypted credentials

DELETE /api/v1/settings/payments/{provider} — disable provider

POST /api/v1/settings/payments/{provider}/test — sandbox/live ping

PATCH /api/v1/settings/payments/defaults — default\_mode, fallback rules

GET /api/v1/settings/payments/webhook-urls — copy-paste for merchant dashboards

At sale time:

GET /api/v1/checkout/payment-methods — returns enabled methods for branch

POST /api/v1/payments/manual-momo — { sale\_id, provider, amount, ref?, msisdn\_last4? }

POST /api/v1/payments/momo/initiate — only if paystack/hubtel configured + online

## J. Payment UI — POS checkout screen

Payment drawer shows only **tenant-enabled** methods:

[ Cash ] [ MoMo — Scan/API ] [ MoMo — Manual ] [ Card ]

If no provider configured: show Cash + Manual MoMo only with banner:

'Connect Paystack or Hubtel in Settings to accept automated MoMo.'

Manual MoMo modal: provider dropdown, amount (pre-filled), reference field, confirm.

Automated MoMo modal: MSISDN input! OTP wait spinner! success/fail.

Offline banner: 'Working offline — automated MoMo unavailable'.

Admin Settings! Payments: toggle cards per provider, test keys, view webhook U

## K. Security — tenant payment secrets

Encrypt config\_encrypted with ENCRYPTION\_MASTER\_KEY (32-byte, in KMS/Vault in prod).

Per-tenant derived key: HKDF(master, tenant\_id) — limits blast radius.

API never logs decrypted secrets; test endpoint uses in-memory only.

RBAC: only HQ admin role can view/edit payment settings.

Audit every change to tenant\_payment\_providers.

Webhook routing: tenant\_slug in URL path !' lookup tenant !' verify signature with

Pen test: cross-tenant payment key access must return 403.

## L. Marketing site integration (SaaS)

oceancyber.net /products/pos CTAs updated:

- **Start free trial** !' https://app.pos.oceancyber.net/signup?utm\_source=website
- **Book a demo** !' contact form (for Enterprise / assisted onboarding)

Marketing site does NOT host POS app — only links out to SaaS app domain.

Optional: embed pricing table with deep links per plan ?plan=growth

Lead tracking: pass utm\_\* query params through signup !' store on tenant for anal

## M. SaaS sprint backlog additions

SAAS-1 Public signup + email verification

SAAS-2 Onboarding wizard UI (5 steps) + API

SAAS-3 OceanCyber Paystack subscription + trial + webhooks

SAAS-4 tenant\_payment\_providers table + encryption service

SAAS-5 Settings !' Payments admin UI (Paystack/Hubtel BYOK)

SAAS-6 Manual MoMo payment flow on POS client

SAAS-7 Payment method gating on checkout (enabled providers only)

SAAS-8 Per-tenant webhook routing + signature verification

SAAS-9 Plan limit enforcement middleware

SAAS-10 past\_due / read\_only tenant states + billing dunning emails

SAAS-11 Self-serve password reset + invite team emails

SAAS-12 Marketing site CTA links + utm capture

## N. Revised sales paths (dual channel)

**Self-serve (primary):** Website !' Sign up !' Trial !' Configure payments !' Go live

Target: SMEs, single-branch retail, cash-heavy shops upgrading from notebook.

**Sales-assisted (secondary):** Website !' Book demo !' OceanCyber provisions + t

Target: Multi-branch, pharmacy, enterprise, merchants needing catalog import help.

Both paths use same tenant model. Assisted path: OceanCyber admin uses impersonate + pre-fill onboarding.

Enterprise custom pricing: manual plan override on tenant record.

## O. Open decisions — updated (v4.0)

OD-6 RESOLVED: Self-serve signup YES — core SaaS model.

OD-8 Trial length: 14 vs 30 days — default 14, A/B later.

OD-9 Require Paystack card on trial start vs trial without card — default: trial without card, read\_only after.

OD-10 OceanCyber MoMo aggregator tier (optional future): OceanCyber collects on behalf — regulatory review required.

OD-11 POS app domain: app.pos.oceancyber.net vs pos.oceancyber.net — pick one, cookie scope.

### 3.1 Marketing spec — checkout

- Barcode scan + SKU search + quick keys (top sellers)
- Line-item and cart-level discounts (supervisor PIN above threshold)
- Tax lines: NHIL/GETFund/VAT configurable per tenant (Ghana retail norms)
- Payments: Cash, Card (terminal phase 2), Automated MoMo (tenant Paystack/Hubtel), Manual MoMo, Telecel/AT
- Change due calculation for cash
- Receipt: 80mm thermal ESC/POS + optional SMS/email
- Shift: open float, close count, variance report
- Hold/recall cart for busy counters

### 3.2 Marketing spec — offline sync

- SQLite on device: catalog snapshot + pending operations queue
- Sale allowed offline up to configurable limit (default 72h sync window)
- Operations: SALE\_CREATE, SALE\_VOID, PAYMENT\_ADD, STOCK\_ADJUST, SHIFT\_CLOSE



- Client sequence: monotonic per (device\_id); server idempotency key = client UUID
- UI: green/yellow/red sync indicator; pending count badge
- On reconnect: batch upload !' server ack !' download catalog delta
- Conflict: duplicate client UUID rejected; stock conflicts !' supervisor queue

### 3.3 Marketing spec — inventory & branches

INVENTORY: SKU, barcode, category, cost, sell price, reorder level, supplier ref.

Stock moves: sale decrement, GRN (goods received), transfer out/in, adjustment with reason.

Alerts: email/SMS to branch manager when below reorder level.

MULTI-BRANCH: branch entity with address, timezone, tax profile.

HQ can push price updates; branch manager can request local override (approval workflow optional).

Dashboards: today vs yesterday, payment mix pie, top 10 SKUs, hourly heatmap.

### 3.4 Marketing spec — restaurant & onboarding

RESTAURANT: floor plans, table states (free/occupied/bill requested), send-to-kitchen, KOT print, split by seat or amount.

MOBILE SUPERVISOR: PWA or RN app — approve voids, view live sales, low stock push.

ONBOARDING PACK (OceanCyber delivery):

1. Discovery form (vertical, branches, payment methods, hardware)
2. Catalog import CSV/Excel template
3. Device pairing session
4. 2h staff training + printed quick-start
5. Launch week WhatsApp support group

## 4. Industry vertical presets

Each preset seeds demo catalog, tax defaults, and report labels:

Retail — barcode-heavy, weigh-scale optional, loyalty card field.

Tourism/hospitality — room charge folio link (phase 3), restaurant tables enabled.

Healthcare/pharmacy — batch/expiry on SKU, controlled discount limits.

Financial/agency — transaction caps, mandatory ID field on sale.

Agriculture — bulk unit (bag, crate) conversions.

Logistics — depot counter, driver float tracking.

Real estate — service SKUs (maintenance fees), invoice-style receipt.

Media/events — timed sessions, combo tickets.

## 5. Pricing tiers & feature gates

Starter GHS 99/mo — 1 branch, 3 users, 1 device, cash+MoMo, offline, basic reports.

Growth GHS 199/mo — 3 branches, 10 users, 5 devices, inventory+PO, SMS receipts.

Multi-branch GHS 299/mo — 10 branches, 25 users, transfers, HQ dashboard, API read access.

Enterprise custom — unlimited, GRA E-VAT module, SLA 99.9%, dedicated CSM, custom integrations.

Hardware kits quoted separately (GHS 800–3,500+): tablet, Bluetooth printer, scanner.

Annual prepay: 10 months price for 12 months.

Enforce limits in API middleware (branch count, device count, user seats).

## 6. Recommended tech stack

Monorepo suggestion: oceancyber-pos/ with apps/pos-desktop, apps/pos-admin, apps/api, packages/shared.

POS desktop: Electron 28+ or Tauri 2 + React 18 + TypeScript + Tailwind.

POS tablet: React Native (shared TS business logic) or responsive PWA for pilot.

Admin portal: Next.js 14 or Vite+React — tenant HQ configuration.

API: NestJS 10 (aligns with OceanCyber team) — REST + Socket.io for live dashboard.

DB: PostgreSQL 16 — multi-tenant row isolation.

Cache/queue: Redis 7 + BullMQ — webhooks, SMS, sync batch processing.

Local: SQLite via better-sqlite3 (desktop) or expo-sqlite (mobile).

Payments: Paystack + Hubtel aggregators first; direct MTN/Telecel APIs at enterprise volume.



SMS: Hubtel or Africa's Talking.

Observability: Sentry, OpenTelemetry, structured JSON logs (pino).

CI: GitHub Actions — lint, test, build artifacts, Electron code sign (Windows/macOS).

## 7. Repository layout (proposed)

```
oceancyber-pos/
  apps/api/           - NestJS multi-tenant API
  apps/admin/         - HQ web portal
  apps/pos-desktop/   - Electron register client
  apps/pos-mobile/    - RN supervisor app (phase 2)
  packages/shared/    - DTOs, enums, sync protocol types
  packages/receipt/   - ESC/POS builder
  prisma/             - schema + migrations
  docker-compose.yml  - postgres, redis, api, admin
  docs/              - OpenAPI spec, this guide
```

Separate Git repo from oceancyber.net marketing monorepo.

## 8. System architecture

Clients !•HTTPS/WSS! API Gateway (TLS, rate limit) ! NestJS modules.

PostgreSQL (primary), Redis (cache, queues, session).

Workers: payment-webhook, sms-send, sync-batch, report-aggregate (nightly cron).

Device boot flow: login ! fetch tenant config ! download catalog snapshot (version every 60s).

Catalog updates: HQ publishes ! increment catalog\_version ! clients pull delta on

Admin portal talks to same API with user JWT (not device token).

Marketing site does NOT call POS API publicly — leads go to oceancyber.net CRM/contact pipeline.

## 9. Database schema — core tenancy

```
tenants — id, name, slug, plan_tier, status, timezone, currency, created_at
branches — id, tenant_id, name, code, address, phone, is_active
users — id, tenant_id, email, password_hash, role, branch_id (nullable for HQ)
devices — id, tenant_id, branch_id, name, device_token_hash, last_seen_at, app_version
subscriptions — id, tenant_id, plan, amount_ghs, paystack_sub_code, renews_at, status
```

All business tables include tenant\_id UUID NOT NULL + index (tenant\_id, ...).

Use Prisma or Drizzle ORM; enforce RLS optional second layer in PostgreSQL.

## 9. Database schema — catalog & inventory

```
categories — id, tenant_id, name, parent_id, sort_order
products — id, tenant_id, sku, barcode, name, category_id, unit, tax_rate_id, is_active
product_prices — id, product_id, branch_id (null=HQ default), price_ghs, effective_from
stock_levels — id, tenant_id, branch_id, product_id, quantity, reserved_qty
stock_movements — id, tenant_id, branch_id, product_id, delta, reason, ref_type, ref_id, user_id
suppliers — id, tenant_id, name, phone, email
purchase_orders — id, tenant_id, branch_id, supplier_id, status, ordered_at
po_lines — id, po_id, product_id, qty_ordered, qty_received, unit_cost
branch_transfers — id, from_branch, to_branch, status, shipped_at, received_at
```

## 9. Database schema — sales & payments

```
shifts — id, tenant_id, branch_id, device_id, user_id, opened_at, closed_at, opening_float, closing_cash
sales — id, tenant_id, branch_id, device_id, shift_id, client_uuid, receipt_no, status, subtotal, tax, total, created_at
sale_lines — id, sale_id, product_id, qty, unit_price, discount, line_total
payments — id, sale_id, method (cash|momo|card|telecell|at), amount, status, external_ref, idempotency_key
momo_transactions — id, payment_id, provider, msisdn_masked, telco_ref, callback_payload_json
voids — id, sale_id, reason, approved_by, voided_at
refunds — id, sale_id, amount, method, approved_by (phase 2)
```

receipt\_no: branch prefix + sequential per branch (e.g. ACC-20260708-0042).

## 9. Database schema — restaurant & sync

tables — id, branch\_id, floor\_id, label, seats, status

floors — id, branch\_id, name, layout\_json

table\_orders — id, table\_id, sale\_id, status (open|kitchen|served|paid)

kitchen\_tickets — id, table\_order\_id, station, items\_json, printed\_at

sync\_batches — id, device\_id, uploaded\_at, op\_count, status

sync\_operations — id, batch\_id, client\_seq, op\_type, payload\_json, server\_applied\_at, conflict\_flag

catalog\_snapshots — id, tenant\_id, version, manifest\_json, created\_at

## 10. REST API — auth & tenancy

POST /api/v1/auth/register — self-serve signup (creates user + tenant)

POST /api/v1/auth/verify-email — activate account

POST /api/v1/auth/login — email/password! user JWT +  
POST /api/v1/auth/device/pair — pairing code from admin

POST /api/v1/auth/refresh — rotate tokens

GET /api/v1/me — user profile + permissions + tenant status

GET /api/v1/tenant — tenant config, plan limits, feature flags, onboarding\_step

Headers: Authorization Bearer (user or device), X-Tenant-Id (optional if embedded in JWT).

Device tokens scoped: branch\_id, device\_id, permissions cashier-only.

## 10. REST API — catalog & sync

GET /api/v1/catalog/snapshot?since\_version=N — full or delta catalog

GET /api/v1/products — paginated search (admin)

POST /api/v1/products — create (admin, branch\_manager+)

PATCH /api/v1/products/:id — update

POST /api/v1/sync/push — body: { device\_id, operations[] }

GET /api/v1/sync/status — pending conflicts for supervisor

POST /api/v1/sync/resolve — approve/reject conflict

## 10. REST API — sales & payments

POST /api/v1/shifts/open — opening float

POST /api/v1/shifts/close — closing count

POST /api/v1/sales — create sale (online) or via sync push (offline)

GET /api/v1/sales/:id — receipt detail

POST /api/v1/sales/:id/void — supervisor approval

POST /api/v1/payments/manual-momo — manual MoMo record (no API)

POST /api/v1/payments/momo/initiate — automated MoMo (tenant's Paystack/Hubtel keys)

GET /api/v1/payments/momo/:id/status — poll

POST /api/v1/webhooks/paystack/{tenant\_slug} — in-store payments (tenant secret)

POST /api/v1/webhooks/hubtel/{tenant\_slug} — in-store payments (tenant secret)

POST /api/v1/billing/webhooks/paystack — OceanCyber SaaS subscription (platform secret)

## 10. REST API — inventory, reports, admin

GET/POST /api/v1/stock/levels — branch stock

POST /api/v1/stock/adjust — reason required

POST /api/v1/purchase-orders — create PO

POST /api/v1/transfers — branch transfer

GET /api/v1/reports/daily?branch\_id&date — sales summary

GET /api/v1/reports/payment-mix — by method

GET /api/v1/reports/top-products — limit 20

GET /api/v1/reports/export.csv — finance export

POST /api/v1/admin/tenants — OceanCyber internal provision

POST /api/v1/admin/tenants/:id/impersonate — support only, audited

## 11. WebSocket events (live dashboard)

Channel: tenant:{id}:branch:{id}

sale.created — { sale\_id, total, method }

payment.momo.pending — { sale\_id, msisdn\_masked }

payment.momo.success | payment.momo.failed

sync.conflict — { device\_id, op\_type }

stock.low — { product\_id, qty, reorder\_level }

Admin dashboard subscribes on login; branch managers see own branch only.

## 12. MoMo payment state machine

States: INITIATED !' PROMPT\_SENT !' AWAITING\_OTP !' SUCCESS | FAILED | TIME

Flow:

1. Cashier selects MoMo, enters customer MSISDN (233XXXXXXXXX)
2. API creates payment row + idempotency\_key (UUID)
3. Call Paystack/Hubtel charge API with callback URL
4. Customer approves on phone; webhook arrives
5. Verify signature; match idempotency\_key; mark sale PAID; print receipt

Timeout: 120s default !' offer retry or switch to cash.

Duplicate webhook: idempotent — return 200 without double-settling.

Store raw callback JSON for disputes (90-day retention minimum).

## 13. Offline sync protocol (detail)

Operation envelope: { client\_uuid, client\_seq, op\_type, payload, device\_timestamp }

Server processing order: sort by client\_seq per device; reject gaps with RESYNC\_REQUIRED.

SALE\_CREATE payload: lines[], payments[], shift\_id, client\_receipt\_no.

Idempotency: client\_uuid unique index — replay returns original server sale\_id.

Catalog delta: { version, products\_added[], products\_updated[], products\_removed[] }.

Device stores catalog\_version locally; skip full download if version match.

Max offline sales without sync: configurable 500 ops or 72h — then block new sales until sync.

## 14. UI screens — POS desktop client

Login / device pairing

Shift open (float entry)

Main sell — search, cart, numpad, payment drawer

Payment — method tabs, MoMo MSISDN, waiting spinner

Receipt preview + print

Shift close — count cash, variance

Supervisor override PIN modal

Sync status drawer

Settings — printer test, logout

Brand UI: dark #0c0c10, accent lime #bbf340, large touch targets (min 44px), high contrast for shop floor.

## 14. UI screens — admin portal

Dashboard — KPI cards, payment mix chart, branch selector

Products — grid CRUD, import CSV, barcode print

Inventory — levels, PO list, transfer wizard

Branches & devices — pairing QR/code

Users & roles — invite email

Reports — date range, export

Settings — tax rates, receipt footer, MoMo credentials (encrypted)

Billing — plan, usage vs limits (phase 2 Paystack sub)

## 15. Security requirements

TLS 1.2+ everywhere; HSTS on admin and API.  
Passwords: bcrypt/argon2; lockout after 5 failures.  
Device tokens: long random, rotatable, stored hashed server-side.  
PII: mask MSISDN in logs (233XX\*\*\*XX12); encrypt MoMo API keys at rest (AES-256).  
RBAC enforced in guards + DB queries scoped by tenant\_id.  
Audit log: voids, price changes, role changes, admin impersonation.  
Webhook signatures mandatory in production — reject unsigned.  
Ghana DPA: privacy policy, data retention config, export/delete on request (enterprise).  
Pen test before enterprise tier launch.

## 16. Environment variables (product)

DATABASE\_URL, REDIS\_URL  
JWT\_SECRET, JWT\_REFRESH\_SECRET, DEVICE\_TOKEN\_PEPPER  
API\_PUBLIC\_URL, ADMIN\_PUBLIC\_URL, CORS\_ORIGINS  
PAYSTACK\_SECRET\_KEY, PAYSTACK\_WEBHOOK\_SECRET  
HUBTEL\_CLIENT\_ID, HUBTEL\_CLIENT\_SECRET, HUBTEL\_WEBHOOK\_SECRET  
SMS\_API\_KEY, SMS\_SENDER\_ID  
SENTRY\_DSN, NODE\_ENV  
TENANT\_DEFAULT\_TIMEZONE=Africa/Accra  
CURRENCY\_DEFAULT=GHS  
OFFLINE\_MAX\_HOURS=72  
MOMO\_TIMEOUT\_SECONDS=120  
OCEANCYBER\_INTERNAL\_API\_KEY — provisioning from CRM (optional)

## 17. Local development setup

```
git clone git@github.com:oceancyber/oceancyber-pos.git
cp .env.example .env
docker compose up -d postgres redis
npm install
npx prisma migrate dev
npm run seed — demo tenant + products
npm run dev:api — http://localhost:4200
npm run dev:admin — http://localhost:4201
npm run dev:desktop — Electron hot reload
```

Demo credentials in seed: demo@merchant.gh / (from SEED\_PASSWORD env).  
Paystack test keys from dashboard; use test MSISDN from Paystack docs.

## 18. Deployment — Coolify VPS

Separate Coolify resource from oceancyber.net marketing stack.

Services: postgres, redis, api (port 4000), admin (port 3000).  
Domains: api.pos.oceancyber.net! api; admin.pos.oceancyber.net! admin.  
Env: all secrets in Coolify UI; never commit .env.  
COMPOSE\_PARALLEL\_LIMIT=1 on 4GB VPS during first build.  
Postgres backups: daily pg\_dump to object storage.  
Desktop clients: download from releases.pos.oceancyber.net or auto-update S3/R2.  
Do NOT merge POS API into marketing docker-compose.yml without architecture review.

## 19. Phase 0 sprint backlog (weeks 1–6)

S0-1 Monorepo scaffold, CI, Docker postgres/redis  
S0-2 Prisma schema: tenants, branches, users, devices  
S0-3 Auth: login, JWT refresh, device pairing code  
S0-4 Products CRUD API + admin UI list/create  
S0-5 Catalog snapshot endpoint + version hash  
S0-6 Electron shell: login, catalog download, empty cart UI

S0-7 Sale create API (cash only) + receipt number seq  
S0-8 Shift open/close API + desktop flows  
S0-9 Thermal print ESC/POS proof of concept  
S0-10 Seed script + internal admin tenant provision CLI

## 20. Phase 1 MVP backlog (weeks 7–16)

S1-1 MoMo initiate + Paystack webhook + polling fallback  
S1-2 SQLite offline queue on desktop + sync push endpoint  
S1-3 Idempotency + conflict detection  
S1-4 SMS receipt via Hubtel  
S1-5 Supervisor PIN void/discount  
S1-6 Daily report API + admin dashboard charts  
S1-7 Stock decrement on sale + basic stock levels  
S1-8 Pilot merchant #1 onboarding (real catalog)  
S1-9 Load test 50 concurrent sales/min per branch  
S1-10 Beta release desktop installer (Windows primary, Ghana market)

## 21. Phase 2 growth backlog (weeks 17–24)

S2-1 Multi-branch HQ dashboard roll-up  
S2-2 Branch transfers + PO workflow  
S2-3 Telecel Cash provider adapter  
S2-4 Restaurant tables + kitchen ticket print  
S2-5 Plan tier enforcement middleware  
S2-6 Paystack subscription billing linkage  
S2-7 CSV export for accountants  
S2-8 Industry preset import CLI  
S2-9 Supervisor mobile PWA  
S2-10 Public marketing site integration — demo request! CRM ticket auto-create

## 22. Phase 3 enterprise backlog

E-1 GRA E-VAT certified integration (legal gate)  
E-2 Card terminal integration (Verifone/Ingenico partner)  
E-3 Direct MTN API at volume tier  
E-4 Advanced analytics — cohort, basket analysis  
E-5 Hardware kit fulfillment partner workflow  
E-6 99.9% SLA monitoring + status page  
E-7 SSO for enterprise HQ (Google/Microsoft)  
E-8 Read-only API for ERP/accounting export

## 23. Testing strategy

Unit: tax calculation, change due, idempotency merge, receipt number format.  
Integration: Supertest on NestJS — sale + payment webhook path.  
E2E: Playwright on admin; Spectron/Playwright on Electron for sell flow.  
MoMo: Paystack sandbox + real device on staging before prod keys.  
Offline: automated test — queue 20 ops offline, sync, assert ledger balance.  
Load: k6 script 100 RPS sync push sustained 5 min.  
UAT: pilot merchant sign-off checklist per vertical preset.

## 24. GRA E-VAT & compliance roadmap

V1: configurable tax components; export CSV labeled for manual GRA filing.  
V2: e-invoice UUID generation internal only (staging).  
V3: GRA-certified submission API after third-party audit — update marketing only when certified.  
Ghana Data Protection Act: register as data controller if >1000 customer records across tenants.  
Receipt retention default 7 years configurable; MSISDN retention 90 days unless loyalty opt-in.  
Terms of service: merchant is controller for end-customer PII; OceanCyber is processor.

## 25. Acquisition & onboarding paths

PATH A — Self-serve SaaS (primary):

1. app.pos.oceancyber.net/signup
2. Verify email! onboarding wizard (business, branch, plan, payments, device)
3. Start trial or subscribe (OceanCyber billing)
4. Configure own Paystack/Hubtel OR use cash + manual MoMo only
5. Go live

PATH B — Sales-assisted (Enterprise / complex):

1. oceancyber.net contact / demo request
2. OceanCyber admin pre-provisions tenant or assists through wizard
3. Same SaaS product — optional white-glove catalog import + training

CRM fields: acquisition\_channel (self\_serve|demo|partner), utm\_source, plan, CSM owner.

## 26. Marketing! engineering alignment

Before each release, verify marketed features:

```
%i Mockup screen exists in product or is marked roadmap
%i Tier gate documented in admin feature flags
%i MoMo tested MTN + Telecel sandbox wallets
%i Offline demo script rehearsed for sales
%i All amounts GHS — no USD in UI
%i GRA claims match certification status
%i Support SLA documented per tier
%i Known limitations listed in demo script (e.g. card t
```

## 27. Competitive differentiation (Ghana 2026)

Table stakes: MoMo, offline, multi-branch, GHS pricing, local support.

OceanCyber wins on:

- White-glove onboarding (agency-grade delivery)
- Security-aware architecture (audit logs, encryption)
- Cross-sell to web/mobile/custom builds for merchants outgrowing POS
- ThinQ/commerce case study credibility

Do not race to lowest price — match GHS 99 entry but sell Growth/Multi-branch on implementation value.

## 28. Open decisions (log before build)

OD-1 Electron vs Tauri for desktop (Tauri = smaller binary, Electron = faster MVP).

OD-2 Paystack vs Hubtel primary MoMo aggregator.

OD-3 Single repo vs api repo + client repo.

OD-4 Card terminal partner and timeline.

OD-5 GRA E-VAT partner/certification body.

OD-6 RESOLVED: Self-serve SaaS signup is required — see sections A–O.

OD-7 Franchise pricing model for resellers.

Record decisions in docs/pos/ADR/ folder as Architecture Decision Records.

## 29. Ghana tax calculation (retail defaults)

Configurable per tenant/branch; defaults reflect common Ghana retail invoicing.

Typical stack (verify with merchant accountant):

- NHIL — 2.5% on taxable amount
- GETFund levy — 2.5% on taxable amount
- COVID-19 Health Recovery Levy — 1% (if still applicable to sector)
- VAT — 15% on (base + levies) for standard-rated goods

Calculation order (implement as pipeline):

1. line\_subtotal = qty × unit\_price " line\_discount
2. cart\_subtotal = sum(line\_subtotal)
3. Apply cart-level discount

4. Compute levies on discounted subtotal per product tax class
5. VAT on levy-inclusive amount where product.tax\_class = STANDARD
6. Zero-rated / exempt products skip VAT but may still need levy rules per advisor

Store tax breakdown on sale row for audit: nhil, getfund, covid, vat, total\_tax.

Round to 2 decimal pesewas (GHS) — round half up per line or per cart (document choice).

### 30. Permission matrix (RBAC)

Resource | Cashier | Supervisor | Branch Mgr | HQ Admin | OceanCyber Support

```

sale.create — ' | ' | ' | ' | '
sale.void — ' | ' | ' | ' | '
discount.above_limit — ' | PIN | ' | ' | '
shift.open_close — ' | ' | ' | ' | '
stock.view — ' | ' | ' | ' | read
stock.adjust — ' | ' | ' | ' | '
po.create — ' | ' | ' | ' | '
transfer.create — ' | ' | ' | ' | '
product.edit — ' | ' | ' | ' | '
user.manage — ' | ' | ' | ' | '
device.pair — ' | ' | ' | ' | '
report.export — ' | ' | ' | ' | read
tenant.settings — ' | ' | ' | ' | ' (impersonate audited)

```

Implement via NestJS guards + permission enum in JWT claims.

### 31. API error codes

Format: { code, message, details?, request\_id }

AUTH\_001 — Invalid credentials  
 AUTH\_002 — Token expired  
 AUTH\_003 — Device not paired  
 AUTH\_004 — Insufficient permissions  
 TENANT\_001 — Tenant suspended  
 TENANT\_002 — Plan limit exceeded (branches/users/devices)  
 SALE\_001 — Shift not open  
 SALE\_002 — Sale already voided  
 SALE\_003 — Duplicate client\_uuid  
 PAY\_001 — MoMo timeout  
 PAY\_002 — MoMo declined  
 PAY\_003 — Idempotency conflict  
 SYNC\_001 — Sequence gap — resync required  
 SYNC\_002 — Offline window exceeded  
 SYNC\_003 — Conflict pending supervisor  
 STOCK\_001 — Insufficient stock  
 STOCK\_002 — Transfer in transit  
 VAL\_001 — Validation error (details.field[])

HTTP: 400 validation, 401 auth, 403 forbidden, 409 conflict, 422 business rule, 429 rate limit.

### 32. Sample JSON — create sale (online)

POST /api/v1/sales

```

{
  "client_uuid": "550e8400-e29b-41d4-a716-446655440000",
  "shift_id": "shift_abc",
  "lines": [{ "product_id": "prod_1", "qty": 2, "unit_price": 45.00 }],
  "payments": [{ "method": "cash", "amount": 95.00 }],
  "discount_total": 0
}

```

Response 201:



```
{
  "id": "sale_xyz", "receipt_no": "ACC-20260708-0042",
  "subtotal": 90.00, "tax": 5.00, "total": 95.00, "status": "completed"
}
```

### 33. Sample JSON — sync push (offline batch)

POST /api/v1/sync/push

```
{
  "device_id": "dev_001",
  "catalog_version_local": 42,
  "operations": [
    {
      "client_uuid": "550e8400-e29b-41d4-a716-446655440000",
      "client_seq": 104,
      "op_type": "SALE_CREATE",
      "device_timestamp": "2026-07-08T14:22:01Z",
      "payload": { /* same as sale create */ }
    }
  ]
}
```

Response 200:

```
{
  "applied": 1, "conflicts": [],
  "catalog_delta": { "version": 43, "products_updated": [] }
}
```

### 34. Sample JSON — MoMo initiate & webhook

POST /api/v1/payments/momo/initiate

```
{ "sale_id": "sale_xyz", "msisdn": "233241234567", "provider": "mtn" }
```

Response 202:

```
{ "payment_id": "pay_1", "status": "PROMPT_SENT", "idempotency_key": "..." }
```

Webhook POST /api/v1/webhooks/paystack (signature header):

```
{ "event": "charge.success", "data": { "reference": "...", "amount": 9500 } }
```

Server: verify HMAC !' find payment by reference !' idempotent complete !' emit sal

### 35. Receipt template (80mm thermal)

Width: 48 chars (font A) or 32 chars (font B large).

Header block:

- Merchant legal name (tenant.trading\_name)
- Branch address + phone
- TIN if configured (GRA prep)
- Receipt # + date/time (Africa/Accra)
- Cashier name + shift #

Body: each line — name (wrap), qty × price, line total

Subtotal, NHIL, GETFund, COVID, VAT, TOTAL (bold)

Payment: MoMo / Cash — amount, MoMo ref if applicable

Footer:

- Thank you message (configurable)
- Return policy snippet
  - QR code !' optional digital receipt URL (phase 2)
- Powered by OceanCyber POS (Starter tier) or white-label (Enterprise)

### 36. SMS receipt template

Max 160 chars single SMS; use 2-part if needed.

**Template:**

```
{merchant}: Receipt {receipt_no}. Total GHS {total}. Paid via {method}. {date}. Queries: {branch_phone}
```

MoMo success adds: Ref {telco\_ref}.

Opt-in: customer\_msisdn only if cashier enters for SMS or loyalty enrolled.

Rate limit: 10 SMS/min per branch — prevent abuse.

## 37. Hardware kit specifications

Starter kit (~GHS 1,200–1,800):

- Android tablet 10" or Windows mini-PC (i3, 8GB RAM, 128GB SSD)
- Bluetooth thermal printer 80mm ESC/POS
- USB barcode scanner (keyboard wedge)
- UPS battery backup 650VA (ECG outage survival)

Growth kit adds: cash drawer RJ11 kick, customer pole display (optional).

Restaurant kit adds: kitchen printer IP/Ethernet, tablet for waiter.

Certify: printer + scanner models in docs/pos/HARDWARE-COMPAT.md before sales quote.

## 38. Supported devices (certify before sale)

Printers: Epson TM-T20III, Xprinter XP-N160II, generic ESC/POS BT.

Scanners: Honeywell Voyager, Zebra DS2208, USB wedge clones.

Tablets: Samsung Galaxy Tab A9+, Lenovo M10 (Android 12+).

Desktop: Windows 10/11 64-bit primary; macOS phase 3.

Test matrix: print 100 receipts, scan 500 barcodes, 8h continuous run.

Document failures in compatibility matrix — do not sell uncertified combos.

## 39. Entity relationships (ERD summary)

tenant 1—N branches 1—N devices

tenant 1—N users (role, optional branch\_id)

branch 1—N shifts 1—N sales 1—N sale\_lines

sale 1—N payments 1—0..1 momo\_transactions

product N—1 category; product 1—N product\_prices (branch override)

branch + product !' stock\_levels; stock\_movements audit all deltas

purchase\_order 1—N po\_lines; branch\_transfer links two branches

floor 1—N tables 1—N table\_orders !' sale

device 1—N sync\_batches 1—N sync\_operations

Every business table: tenant\_id FK + composite indexes for list queries.

## 40. Database indexes (performance)

sales (tenant\_id, branch\_id, created\_at DESC)

sales (tenant\_id, client\_uuid) UNIQUE

payments (idempotency\_key) UNIQUE

products (tenant\_id, sku) UNIQUE

products (tenant\_id, barcode) WHERE barcode IS NOT NULL

stock\_levels (tenant\_id, branch\_id, product\_id) UNIQUE

sync\_operations (device\_id, client\_seq) UNIQUE

users (tenant\_id, email) UNIQUE

Partition consideration: sales by month at >10M rows/tenant (enterprise only).

Nightly job: refresh materialized view branch\_daily\_summary.

## 41. Caching & rate limits

Redis cache:

- catalog\_snapshot:{tenant\_id}:{version} — TTL 1h
- tenant\_config:{tenant\_id} — TTL 5m, invalidate on settings change
- report:daily:{branch}:{date} — TTL until midnight Accra

#### Rate limits (per tenant):

- API general: 600 req/min
- sync push: 120 req/min per device
- MoMo initiate: 30 req/min per branch
- Login: 10 req/min per IP

Return 429 with Retry-After header.

## 42. CI/CD pipeline

GitHub Actions on push to main:

1. lint + typecheck (api, admin, desktop, shared)
2. unit tests (vitest/jest)
3. integration tests (testcontainers postgres)
4. build api Docker image !' push ghcr.io/oceancyber/pos-api:sha
5. build admin static !' artifact
6. build Electron Windows installer (main only, tagged releases)

Staging: auto-deploy api+admin to staging Coolify on main merge.

Production: manual approval gate; tag semver v1.2.3.

Desktop: upload to R2 releases bucket; manifest.json for auto-update.

## 43. Versioning & release policy

API: semver in URL /api/v1/ — breaking changes only in v2.

Desktop client: semver MAJOR.MINOR.PATCH — must match min\_api\_version from /tenant config.

Force upgrade: if client < min\_version, block sell screen with update prompt.

Changelog: docs/pos/CHANGELOG.md per release.

Pilot merchants: staged rollout 10% !' 50% !' 100% over 1 week.

## 44. Electron auto-update flow

Use electron-updater + generic provider (R2/S3).

On launch: check manifest !' download delta in background !' prompt restart.

Critical security patch: force restart after shift close only.

Offline: skip check; retry next online boot.

Sign Windows builds with code signing cert (EV recommended for SmartScreen).

## 45. Backup & disaster recovery

PostgreSQL: daily full pg\_dump 02:00 UTC !' encrypted S3; WAL archiving for PIT

Redis: RDB snapshots every 6h (queue recovery acceptable 6h loss max).

RPO: 24h (Starter), 1h (Enterprise with WAL).

RTO: 4h api restore, 24h full stack.

Runbook: docs/pos/runbooks/DR-RESTORE.md — quarterly drill on staging.

Tenant export: admin button export all sales CSV + catalog JSON (GDPR/DPA request).

## 46. Monitoring & alerting

Metrics (Prometheus/Grafana or Coolify metrics):

- api\_request\_duration\_p95, api\_error\_rate
- momo\_webhook\_lag\_seconds
- sync\_batch\_processing\_time
- devices\_online\_count per tenant

Alerts (PagerDuty/WhatsApp ops):

- API error rate > 2% for 5 min
- MoMo webhook queue depth > 100
- Postgres disk > 80%
- Certificate expiry < 14 days

Status page: status.pos.oceancyber.net (Enterprise SLA public).

## 47. Performance targets (SLA)

API p95 latency: < 300ms read, < 800ms sale create (online).

Sync push 50 ops batch: < 2s p95.

Catalog snapshot cold: < 5s for 5,000 SKUs.

Desktop UI: add-to-cart < 100ms local.

MoMo prompt to customer phone: < 8s p95 (provider dependent).

Uptime: 99.5% Starter, 99.9% Enterprise (excl. planned maintenance).

## 48. Catalog CSV import format

Required columns: sku, name, price\_ghs, category

Optional: barcode, cost\_ghs, tax\_class, unit, reorder\_level, supplier\_code

Example row:

SKU001,Peak Milk 400g,12.50,Dairy,623456789012,10.00,STANDARD,each,20,

Import CLI: npx pos-cli catalog import --tenant=slug --file=products.csv

Validation: duplicate SKU fails row; partial import rolls back unless --force.

Max 10,000 rows per import; split larger catalogs.

## 49. Merchant onboarding checklist

```
%i Signed agreement + plan tier confirmed
%i TIN and legal trading name collected
%i Branch addresses + phones entered
%i Tax profile configured with accountant sign-off
%i MoMo merchant account linked (Paystack subaccount or
%i Catalog CSV imported and spot-checked (10 SKUs)
%i Hardware delivered and tested print/scan
%i Devices paired; cashier accounts created
%i Training session completed (attendance sheet)
%i Pilot day shadow support scheduled
%i Go-live date + launch week support group created
```

## 50. OceanCyber internal provision checklist

```
%i Create tenant in admin CLI with plan + limits
%i Apply industry preset (retail, pharmacy, etc.)
%i Generate device pairing codes (N = register count)
%i Staging demo verified within 24h of prod provision
%i CRM ticket linked (lead source, CSM owner)
%i Billing record: manual invoice or Paystack sub id
%i Document custom config in tenant.notes
%i Add to pilot monitoring dashboard
```

## 51. Marketing site! SaaS app integration

oceancyber.net /products/pos primary CTA: Start free trial! app.pos.oceancyber.  
Secondary CTA: Book demo! contact form (Enterprise / assisted onboarding).

Pass utm\_source, utm\_campaign, plan query params to signup! store on tenant.

POST contact API (marketing backend) still captures demo leads — does NOT create POS tenant.

Self-serve signups create tenant automatically via POST /api/v1/auth/register on POS API.

Calendly optional after demo request on marketing site only.

## 52. Dual billing architecture

BILLING A — SaaS subscription (OceanCyber! merchant):

Platform Paystack account; plans pos\_starter/growth/multibranch monthly.

Webhook: /api/v1/billing/webhooks/paystack — subscription lifecycle.

Trial! active! past\_due! suspended! cancelled states on tenant.

BILLING B — In-store customer payments (merchant! their customers):

Merchant's own Paystack/Hubtel keys in tenant\_payment\_providers.

Webhook: /api/v1/webhooks/paystack/{tenant\_slug} — per-tenant signature.

OceanCyber never holds merchant customer MoMo funds in BYOK model.

Merchants on cash/manual MoMo only: Billing B not configured — Billing A still required for SaaS access.

### 53. Branch transfer workflow

1. Branch manager creates transfer request (from !' to, lines[])
  2. Status: DRAFT !' APPROVED (HQ or auto if same region) !' SHIPPED !' RECEIVE
  3. SHIPPED: decrement from\_branch stock (reserved)
  4. RECEIVED: increment to\_branch; receiver confirms qty variance
  5. Variance > 2% triggers supervisor investigation flag
- Audit: stock\_movements rows for each state transition.

### 54. Restaurant order lifecycle

Table FREE !' order opened !' items sent to kitchen (KOT print) !'  
IN\_KITCHEN !' READY !' SERVED !' BILL\_REQUESTED !' payment !' PAID !' table FRE

Split bill: divide sale\_lines across sub-sales or payment splits.

Merge tables: combine open orders before payment (supervisor).

Kitchen display system (KDS) — phase 3; MVP uses ticket printer only.

### 55. Shift reconciliation procedure

Open: count opening float !' record in system !' unlock sell screen.

During shift: all sales tied to shift\_id + device\_id.

Close: system expects cash = opening + cash\_sales " cash\_refunds.

Cashier enters counted cash; variance > GHS 5 requires supervisor note.

MoMo/card: reconcile against payment report (no manual count).

Close generates Z-report PDF — email to branch manager nightly.

### 56. Void & refund policy matrix

Same-day void (before close): supervisor PIN, stock restored, audit reason.

Next-day void: branch manager only; finance notification.

MoMo void: follow aggregator reversal API if same day; else manual refund process.

Partial refund: phase 2 — new payment row negative amount.

No delete of sale rows — status = voided only (immutable ledger).

### 57. Fraud & abuse controls

Supervisor PIN: 6 digit, lockout 5 fails, reset by branch manager.

Discount max 20% cashier; above requires supervisor.

MoMo: block same MSISDN > 5 transactions/min (velocity check).

Device fingerprint: alert if device\_token used from new IP country.

Audit export for tenant: all voids > GHS 100 in date range.

Employee sale tracking: optional cashier\_id on every line for commission.

### 58. Support tiers & SLA

Starter: email/WhatsApp business hours (Mon–Sat 8–6 Accra), 24h first response.

Growth: same + 4h response for P1 (cannot sell).

Multi-branch: dedicated line, 2h P1 response.

Enterprise: named CSM, 1h P1, 99.9% uptime credit.

P1: system down, MoMo completely blocked, data loss risk.

P2: single device, report wrong, printer failure workaround exists.

P3: feature request, training question.

### 59. Incident response runbook

1. Detect alert or merchant report !' create incident channel
2. Classify P1/P2/P3; assign incident commander
3. P1 MoMo outage: switch to cash-only banner on admin; notify all active tenants
4. Communicate ETA on status page every 30 min

5. Post-mortem within 48h: timeline, root cause, action items

6. Regulated tenants: breach notify if PII exposed per DPA timelines

## 60. Data retention schedule

Sales & payments: 7 years default (tax audit).

MoMo callback JSON: 90 days hot storage !' 7 years cold archive.

MSISDN on receipt: 90 days unless loyalty opt-in (then until opt-out).

Audit logs: 2 years online, 5 years archive.

Deleted tenant: 30-day soft delete !' anonymize PII !' hard delete data at 90 days.

## 61. Penetration test scope (pre-enterprise)

OWASP API Top 10 on /api/v1

JWT tampering, tenant\_id horizontal privilege escalation

Webhook signature bypass attempts

Device token replay

Admin impersonation audit trail integrity

SQL injection on search endpoints

Deliverable: report + remediate critical before Enterprise launch.

## 62. Team roles & ownership

Product owner: OceanCyber founder/ops — backlog priority

Tech lead: architecture, API review, security sign-off

Backend (1–2): NestJS API, sync, payments

Frontend (1): admin portal + Electron POS UI

DevOps (0.5): Coolify, CI, monitoring

QA (0.5): test plans, pilot UAT

CSM: onboarding, training, launch week (non-dev)

Estimated MVP team: 3 engineers × 4 months + CSM part-time.

## 63. Budget & timeline summary

Phase 0–1 MVP: ~16 weeks, 3 FTE engineers

Infra: VPS GHS ~500–800/mo staging+prod, Paystack/Hubtel fees per txn

Code signing cert: ~\$200–400/yr Windows

SMS: ~GHS 0.03–0.05 per receipt

Pilot: 2 merchants free 60 days !' case study

Public launch: after 2 successful pilots + MoMo reconciliation audit

Revenue target illustration: 50 Growth tenants × GHS 199 = GHS 9,950 MRR.

## 64. Glossary

KOT — Kitchen Order Ticket

GRN — Goods Received Note

MSISDN — Phone number in international format (233...)

MoMo — Mobile Money (MTN, Telecel, AT)

TIN — Tax Identification Number (GRA)

E-VAT — Electronic VAT invoicing (GRA Act 1151)

SKU — Stock Keeping Unit

Z-report — End-of-shift sales summary

Tenant — One merchant organization in multi-tenant SaaS

Idempotency — Same request twice produces same result once

## 65. Document revision history

v1.0 — Initial POS dev guide from marketing spec

v2.0 — Expanded: schema, API, sprints, sync protocol

v3.0 — Extended: tax, RBAC, JSON samples, hardware, DR, SLA, runbooks

v4.0 — Complete SaaS: self-serve registration, BYOK payments, offline/manual MoMo, dual billing

Next planned additions: OpenAPI YAML export, Figma wireframe links, ADR decisions.

```
Maintain: edit docs/pos/pos-guide-sections*.mjs !' npm run docs:pos-pdf
```

## P. Wireframe — Sign up & sign in

Screen: /signup (app.pos.oceancyber.net)

Layout: split — left brand panel (lime accent, product mockup); right form.

Fields: Full name, Work email, Phone (+233 prefix), Password + strength meter, Business name.

Checkbox: I agree to Terms, Privacy, and Data Processing Agreement.

Primary CTA: Create account !' loading !' Check your email state.

Screen: /signin

Fields: Email, Password. Links: Forgot password, Create account.

Post-login routing: onboarding incomplete !' /onboarding; else !' /dashboard.

Screen: /verify-email

States: success (auto-redirect onboarding), expired token (resend button), invalid (support link).

Mobile: single column; sticky CTA; min 44px touch targets; dark theme #0c0c10, accent #bbf340.

## P. Wireframe — Onboarding wizard (steps 1–3)

Top: progress bar 5 steps — Business | Branch | Plan | Payments | Team & device.

Step 1 — Business: trading name, industry dropdown (8 verticals), TIN optional, logo upload optional.

Step 2 — Branch: branch name, address, city (Accra default), phone, receipt footer textarea.

Step 3 — Plan: 3 pricing cards (Starter/Growth/Multi-branch), feature comparison table.

Toggle: Start 14-day free trial (Starter) vs Subscribe now.

OceanCyber billing note: You will pay OceanCyber for software – separate from store payments.

Nav: Back | Continue. Skip link on steps 4–5 only.

Save draft on blur — PATCH onboarding/step/N.

## P. Wireframe — Onboarding payments (step 4)

Headline: How will you accept payments in your store?

Option cards (multi-select):

☒ Cash — Always enabled

☐ Automated MoMo (Paystack) — Connect your account

☐ Automated MoMo (Hubtel) — Connect your account

☒ Manual MoMo — Record payments without API (good for offline)

☐ Card terminal — Coming soon

Expand Paystack: Public key, Secret key fields, Test connection button, webhook URL copy box.

Expand Hubtel: Client ID, Client secret, Merchant ID, Test connection.

Banner: Skip for now — start with cash and manual MoMo only. Configure automated MoMo later in Settings.

Security note: Keys encrypted; OceanCyber staff cannot see your secret keys.

## P. Wireframe — Settings !' Payments (HQ admin)

Path: /settings/payments

Section 1 — Default checkout mode: radio Cash | Manual MoMo | Ask every sale.

Section 2 — Provider cards (grid):

Paystack [Enabled/Disabled] [Configure] [Test] Last tested: timestamp

Hubtel [...], Manual MoMo [Always on], Cash [Always on]

Section 3 — Fallback rules:

☒ If automated MoMo fails, offer manual MoMo

☒ If offline, hide automated MoMo

Supervisor PIN required for manual MoMo above GHS [500]

Section 4 — Webhook URLs (read-only, copy buttons per provider)

Section 5 — Audit log table: user, action, provider, timestamp

## P. Wireframe — POS checkout payment drawer

Trigger: Pay button on cart (desktop register, full-width bottom sheet on tablet).



Total due: GHS 127.50 (large). Split payment toggle (phase 2).

Method tiles (only enabled methods shown):

[ Cash ] [ MoMo API ] [ MoMo Manual ] [ Card greyed ]

Cash flow: Amount tendered numpad! Change due! Complete! Print.

MoMo API flow: MSISDN! Confirm! Waiting for customer... spinner! Success/Fail

Manual MoMo: Provider select! Reference optional! Confirm (no spinner).

Offline strip (yellow): Offline — automated MoMo unavailable. Cash and manual MoMo OK.

Past\_due strip (red, tenant billing): Subscription inactive — view-only unless admin renews.

## Q. Prisma schema — tenancy & SaaS billing

```
model Tenant {
  id          String   @id @default(uuid())
  slug        String   @unique
  name        String
  status      TenantStatus @default(PENDING_VERIFICATION)
  planCode    String?
  trialEndsAt DateTime?
  onboardingStep Int @default(1)
  paystackCustomerId String?
  paystackSubCode String?
  industry    String?
  utmSource   String?
  createdAt   DateTime @default(now())
  users       User[]
  branches    Branch[]
  paymentProviders TenantPaymentProvider[]
}

enum TenantStatus {
  PENDING_VERIFICATION TRIALING ACTIVE PAST_DUE SUSPENDED CANCELLED
}
```

## Q. Prisma schema — users & payment providers

```
model User {
  id          String @id @default(uuid())
  tenantId    String
  tenant      Tenant @relation(fields: [tenantId], references: [id])
  email       String
  passwordHash String
  name        String
  role        UserRole @default(HQ_ADMIN)
  emailVerifiedAt DateTime?
  @@unique([tenantId, email])
}

model TenantPaymentProvider {
  id          String @id @default(uuid())
  tenantId    String
  provider     PaymentProvider
  isEnabled    Boolean @default(false)
  configEncrypted Bytes
  webhookSecretEnc Bytes?
  lastTestedAt DateTime?
  testStatus   String?
  @@unique([tenantId, provider])
}

enum PaymentProvider { CASH MANUAL_MOMO PAYSTACK HUBTEL CARD }
```

## Q. Prisma schema — sales & payments (SaaS fields)

```
model Payment {
  id          String @id @default(uuid())
  saleId      String
  method      PaymentMethod
  captureType CaptureType // AUTOMATED | MANUAL | CASH
  amount      Decimal @db.Decimal(12,2)
  status      PaymentStatus
  idempotencyKey String? @unique
  externalRef String?
  manualProvider String? // mtn | telecel | at
  msisdnLast4 String?
  createdAt   DateTime @default(now())
}
```

```
enum CaptureType { AUTOMATED MANUAL CASH }
```

```
enum PaymentMethod { CASH MOMO CARD }
```

```
model AuditPaymentConfig {
  id, tenantId, userId, provider, action, ip, createdAt
}
```

## R. OpenAPI — auth & registration (v1)

```
POST /auth/register
  body: RegisterDto { name, email, phone, password, businessName, acceptTerms }
  201: { userId, tenantId, slug, verificationSent }
  409: email already registered

POST /auth/verify-email { token: string }
POST /auth/resend-verification { email }
POST /auth/login { email, password } !' { accessToken,
POST /auth/forgot-password { email }
POST /auth/reset-password { token, newPassword }
POST /auth/refresh { refreshToken }
POST /auth/logout
```

## R. OpenAPI — onboarding & billing

```
GET /onboarding !' { step, data, completed }
PATCH /onboarding/step/{1-5} body: partial step payload
POST /onboarding/complete !' 200 | 422 missing fields

GET /billing/status !' { plan, status, trialEndsAt, re
POST /billing/start-trial !' requires verified email
POST /billing/subscribe { planCode, authorizationCode? }
POST /billing/change-plan { planCode }
POST /billing/cancel { atPeriodEnd: boolean }
GET /billing/invoices – list PDF links
POST /billing/webhooks/paystack – platform secret, no auth header
```

## R. OpenAPI — payment settings & checkout

```
GET /settings/payments
PUT /settings/payments/{provider} body: { config, isEnabled }
DELETE /settings/payments/{provider}
POST /settings/payments/{provider}/test !' { ok, message, latencyMs }
PATCH /settings/payments/defaults { defaultMode, fallbackRules }
GET /settings/payments/webhook-urls

GET /checkout/payment-methods?branchId= – enabled methods for POS
POST /payments/cash { saleId, tendered }
```

```
POST /payments/manual-momo { saleId, provider, amount, externalRef?, msisdnLast4? }
```

```
POST /payments/momo/initiate { saleId, msisdn, provider } – requires BYOK + online
```

```
POST /webhooks/paystack/{tenantSlug} – tenant webhook secret
```

```
POST /webhooks/hubtel/{tenantSlug}
```

## S. Sample JSON — self-serve registration

```
POST /api/v1/auth/register
```

```
{
  "name": "Ama Mensah",
  "email": "ama@sunshine-retail.gh",
  "phone": "233241234567",
  "password": "SecurePass123!",
  "businessName": "Sunshine Retail Accra",
  "acceptTerms": true,
  "utmSource": "website",
  "planInterest": "starter"
}
```

201 Response:

```
{ "tenantId": "ten_...", "slug": "sunshine-retail-accra", "verificationSent": true }
```

## S. Sample JSON — save Paystack BYOK config

```
PUT /api/v1/settings/payments/paystack
```

```
{
  "isEnabled": true,
  "config": {
    "publicKey": "pk_live_xxxx",
    "secretKey": "sk_live_xxxx",
    "mode": "live"
  }
}
```

200 Response (secrets never returned):

```
{ "provider": "paystack", "isEnabled": true, "publicKeyPreview": "pk_live_...xxx", "lastTestedAt": null }
```

```
POST /settings/payments/paystack/test! { ok: true, la
```

## S. Sample JSON — manual MoMo payment

```
POST /api/v1/payments/manual-momo
```

```
{
  "saleId": "sale_xyz",
  "provider": "mtn",
  "amount": 127.50,
  "externalRef": "MP260708.1234.A56789",
  "msisdnLast4": "4567"
}
```

201 Response:

```
{ "paymentId": "pay_m1", "captureType": "MANUAL", "status": "completed" }
```

Receipt line: MoMo (MTN) — manual — Ref MP260708...

## T. Tenant lifecycle state machine

```
PENDING_VERIFICATION! (verify email)! TRIALING | ACTIVE
TRIALING! (trial ends + no sub)! PAST_DUE (read-only POS)
TRIALING! (subscribe)! ACTIVE
ACTIVE! (payment fails)! PAST_DUE
PAST_DUE! (pay)! ACTIVE | (30d)! SUSPENDED
SUSPENDED! (pay)! ACTIVE | (90d)! CANCELLED
```

**CANCELLED !' data retained 90d !' anonymized**

PAST\_DUE behavior: admin login OK; new sales blocked on POS; reports export allowed.

SUSPENDED: login OK billing page only; POS devices show renewal banner.

Email triggers: trial\_day\_7, trial\_expired, payment\_failed, suspended.

## U. Email templates (transactional)

verify\_email — subject: Confirm your OceanCyber POS account — CTA link 15 min TTL

welcome — after verify — link to /onboarding

trial\_reminder — day 7 and day 12 of 14 — subscribe CTA

trial\_expired — POS read-only — billing link

payment\_failed — retry Paystack link

subscription\_receipt — monthly invoice PDF attachment

team\_invite — invite user to tenant with role

password\_reset — token link 1h TTL

Sender: noreply@pos.oceancyber.net · Reply-to: support@oceancyber.net

Provider: Resend or SendGrid; templates in apps/api/templates/email/

## V. 16-week delivery plan (story points)

Weeks 1–2 (21 pts): monorepo, Prisma tenancy, register+verify API, signup UI

Weeks 3–4 (24 pts): onboarding wizard API+UI, plan selection, trial flag on tenant

Weeks 5–6 (26 pts): OceanCyber billing Paystack sub, webhooks, past\_due middleware

Weeks 7–8 (28 pts): tenant\_payment\_providers encryption, settings UI, test connection

Weeks 9–10 (30 pts): catalog, cash sale, shift, Electron shell, manual MoMo flow

Weeks 11–12 (28 pts): Paystack BYOK MoMo initiate + tenant webhook, offline sync

Weeks 13–14 (24 pts): inventory, daily reports, SMS receipt, beta 5 tenants

Weeks 15–16 (20 pts): multi-branch basics, hardening, public launch

Total ~201 story points · velocity ~12–15 pts/week with 3 engineers.

## W. Environment variables (v5 complete list)

APP\_PUBLIC\_URL=https://app.pos.oceancyber.net

API\_PUBLIC\_URL=https://api.pos.oceancyber.net

DATABASE\_URL, REDIS\_URL

JWT\_SECRET, JWT\_REFRESH\_SECRET, DEVICE\_TOKEN\_PEPPER

ENCRYPTION\_MASTER\_KEY — 32-byte hex for tenant payment secrets

OCEANCYBER\_PAYSTACK\_SECRET\_KEY — SaaS subscription billing only

OCEANCYBER\_PAYSTACK\_WEBHOOK\_SECRET

OCEANCYBER\_PAYSTACK\_PLAN\_STARTER/GROWTH/MULTIBRANCH

EMAIL\_PROVIDER\_API\_KEY, EMAIL\_FROM

CAPTCHA\_SECRET — signup bot protection

SENTRY\_DSN, NODE\_ENV=production

Note: merchant Paystack keys live in DB per tenant — NOT in env.

## X. QA checklist — SaaS signup to first sale

```
%i Register new account !' verification email received
%i Verify email !' lands on onboarding step 1
%i Complete wizard skipping payments !' cash-only enable
%i Pair device !' open shift !' add product !' cash sale !'
%i Return to settings !' add Paystack test keys !' test c
%i MoMo API sale online !' OTP on test phone !' receipt s
%i Toggle airplane mode !' manual MoMo sale !' sync !' app
%i Trial expiry simulation !' POS blocks new sales !' bil
%i Second tenant cannot read first tenant payment confi
%i Webhook with wrong tenant slug signature !' 401
```

## Y. Document revision — v5.0

v5.0 adds: wireframes (signup, onboarding, payments, checkout),

Prisma schema for SaaS tenancy and BYOK payments,  
OpenAPI catalog for auth/onboarding/billing/payment settings,  
JSON samples for register, BYOK, manual MoMo,  
tenant lifecycle state machine, email templates, 16-week sprint plan,  
complete env var list, SaaS QA checklist.

Next: OpenAPI YAML file export, Figma links, ADR-001 Electron choice.

```
Edit docs/pos/pos-guide-sections-v5.mjs !' npm run docs:pos-pdf
```